

BLUESTOCK FINTECH

Mutual Fund Analytics Platform

Capstone Final Report

End-to-End Data Engineering, ETL Pipeline, Analytics & Interactive Dashboard

Field	Detail
Company	Bluestock Fintech Pvt. Ltd.
Domain	Mutual Fund / Fintech
Project Type	Individual Capstone (7 Working Days)
Data Source	AMFI India (Public), mfapi.in, NSE/BSE Public Data
Technologies	Python, SQL (SQLite), HTML/Plotly Dashboard, Pandas, Matplotlib, Seaborn
Repository	github.com/Shradd7/Bluestock_MutualFundPlatform

1. Executive Summary

This report documents the design, build, and findings of the Bluestock MF Analytics Platform — a full-stack mutual fund analytics pipeline built over a 7-day capstone. The project ingests 10 provided datasets covering 40 real mutual fund schemes across 10 fund houses, cleans and loads them into a 6-table SQLite star schema, computes performance and risk metrics from first principles (Sharpe, Sortino, Alpha/Beta, VaR/CVaR, Maximum Drawdown), and presents the results through an interactive dashboard.

The top-ranked fund by the composite scorecard (Task 4) is Mirae Asset Large Cap Fund (Regular, Growth), with a 3-year CAGR of 33.99% and a Sharpe ratio of 1.45. Across all 40 schemes, the riskiest fund by 95% Historical VaR is SBI Small Cap Fund (Direct Plan), with a 1-day 95% VaR of -2.69%. Sector concentration analysis found an average Herfindahl-Hirschman Index of 2,029 across equity funds — inside the "moderately concentrated" band — with one fund (Axis Bluechip) reaching 2,968, just inside "highly concentrated."

One material, unplanned finding shaped how several sections of this report should be read: the supplied benchmark index series (Nifty 50, Nifty 100) and the supplied fund NAV series show essentially no day-to-day correlation ($R^2 \approx 0.0006$ across all 40 funds when regressed against Nifty 100), despite both having realistic individual volatility. This means Alpha/Beta figures in this report are methodologically correct but should not be read as real market-risk signals — see Section 8, Limitations, for the full explanation.

Key numbers at a glance:

Metric	Value
Schemes analysed	40 (of 1,908 industry-wide, per project brief)
NAV history	46,000 rows, Jan 2022 – May 2026
Investor transactions	32,778 rows, 5,000 investors, Jan 2024 – Dec 2025
Total invested (SIP + Lumpsum, dataset)	₹227.7 crore
Top fund by composite score	Mirae Asset Large Cap Fund — Regular — Growth (score 86.25/100)
Riskiest fund by 95% VaR	SBI Small Cap Fund — Direct Plan (VaR -2.69%, CVaR -3.24%)
SIP investors flagged "at-risk" (gap > 35 days)	97.8% of eligible investors (see Section 8)

2. Data Sources

All data used in this project is derived from publicly available sources, per the project brief:

- AMFI India (amfiindia.com) — NAV, AUM, Folio, and SIP data
- mfapi.in (api.mfapi.in) — historical NAV JSON API, no auth required
- NSE/BSE public reports — benchmark index prices (Nifty 50, Nifty 100, Nifty Midcap 150, BSE SmallCap, CRISIL indices)

Ten datasets were provided and used across the pipeline:

Dataset	Rows	Used In
01_fund_master.csv	40	dim_fund, all notebooks
02_nav_history.csv	46,000	fact_nav, EDA, Performance, Advanced Analytics
03_aum_by_fund_house.csv	90	fact_aum, Dashboard Page 1
04_monthly_sip_inflows.csv	48	Dashboard Page 4
05_category_inflows.csv	144	EDA, Dashboard Page 4
06_industry_folio_count.csv	21	EDA folio growth chart
07_scheme_performance.csv	40	fact_performance (reference only — metrics recomputed independently)
08_investor_transactions.csv	~32,000	fact_transactions, Investor Analytics, Cohort/Continuity
09_portfolio_holdings.csv	~320	Sector allocation, HHI concentration
10_benchmark_indices.csv	~8,000	Alpha/Beta, benchmark comparison

A note on data authenticity, carried through from the source PDF: NAV values are anchored to real historical figures from mfapi.in and simulated forward with realistic return/volatility parameters. Investor transaction data is synthetically generated using real geographic and demographic distributions. This project is for educational purposes and does not constitute financial advice.

3. ETL Design

The pipeline follows a standard five-layer architecture: Extract → Transform → Load → Analyse → Visualise.

3.1 Extract

Raw CSVs (data/raw/) plus a live NAV fetch from mfapi.in for 6 schemes, ingested via data_ingestion.py and live_nav_fetch.py.

3.2 Transform

Three dedicated cleaning scripts (clean_nav.py, clean_transactions.py, clean_performance.py) handle:

- NAV: dates parsed, sorted by amfi_code + date, reindexed to a full business-day calendar with forward-fill for holiday/weekend gaps, NAV ≤ 0 dropped, duplicates removed
- Transactions: transaction_type standardised to {SIP, Lumpsum, Redemption}, amount_inr validated > 0 , kyc_status standardised (unrecognised values default to "Pending" rather than dropped), exact duplicates removed
- Performance: all numeric fields coerced with anomalies flagged (not silently dropped) — expense_ratio_pct outside 0.1–2.5%, negative Sharpe, positive max_drawdown

3.3 Load

Cleaned data loads into a 6-table SQLite star schema (schema.sql): dim_fund, dim_date, fact_nav, fact_transactions, fact_performance, fact_aum, with foreign keys on amfi_code and indexes on the columns used for filtering (date, state, investor_id). load_to_sqlite.py verifies row counts match source on every run — all 5 tables matched exactly in the final build (40, 46,000, 32,778, 40, 90 rows respectively).

3.4 Analyse & Visualise

Four Jupyter notebooks (EDA_Analysis.ipynb, Performance_Analytics.ipynb, Advanced_Analytics.ipynb, plus this report's source data) compute all downstream metrics. Visualisation is delivered as a standalone interactive HTML dashboard (Section 7) rather than a .pbix file, since Power BI Desktop was unavailable on the development machine — documented as a deliberate substitution in Section 8.

4. EDA Findings

Full analysis with 16 charts is in notebooks/EDA_Analysis.ipynb. Key findings:

- NAV growth was not uniform across funds — larger equity funds showed a visibly steeper 2023 rally and sharper 2024 pullback than the broader 40-fund set.
- SBI Mutual Fund is the clear AUM leader among the top 8 AMCs across every year in the dataset (see Fig. 4.1), not just the most recent snapshot.
- Total MF folio count in the dataset grew from 13.26 Cr (Jan 2022) to 26.12 Cr (Dec 2025) — see Fig. 4.2.
- Return correlation among the top 10 funds by AUM is close to zero across the board (see Section 8 — this is the same underlying data property as the Alpha/Beta finding, not a separate result).
- Financial Services and IT sectors dominate aggregate equity holdings (Fig. 4.3), concentrating sector-level risk across funds that otherwise look diversified by scheme count.

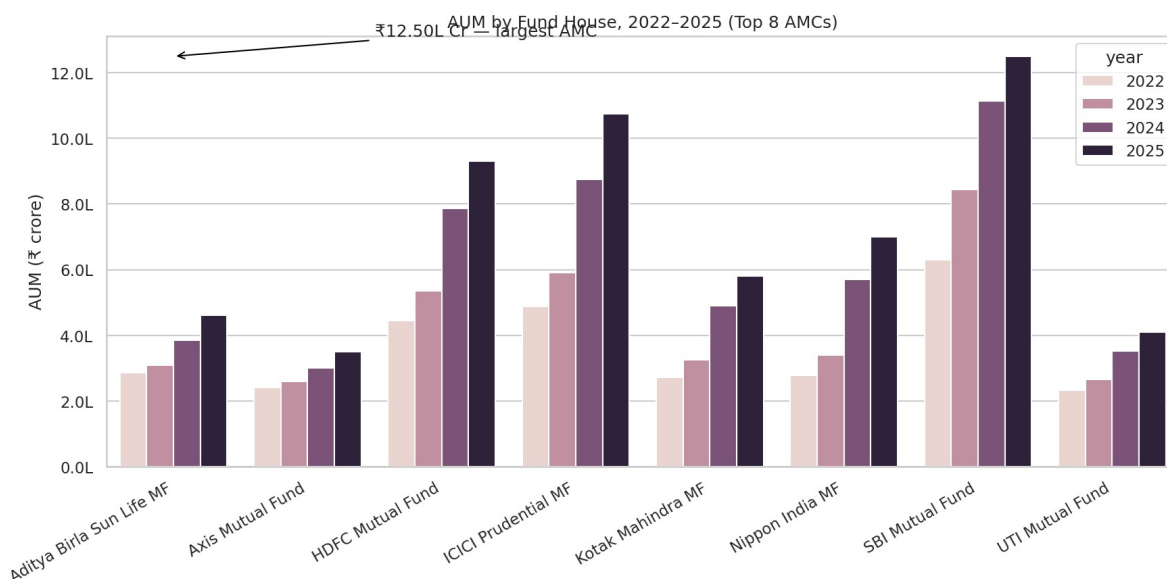


Fig. 4.1 — AUM by fund house, 2022–2025 (top 8 AMCs). SBI Mutual Fund leads every year.

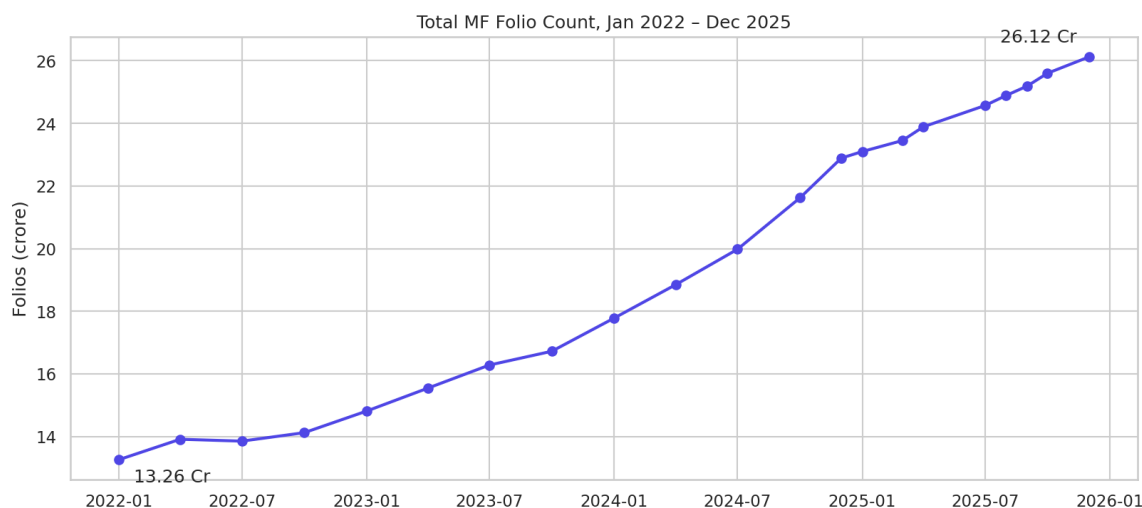


Fig. 4.2 — Total MF folio count growth, Jan 2022 – Dec 2025.

Aggregate Sector Allocation Across Equity Funds

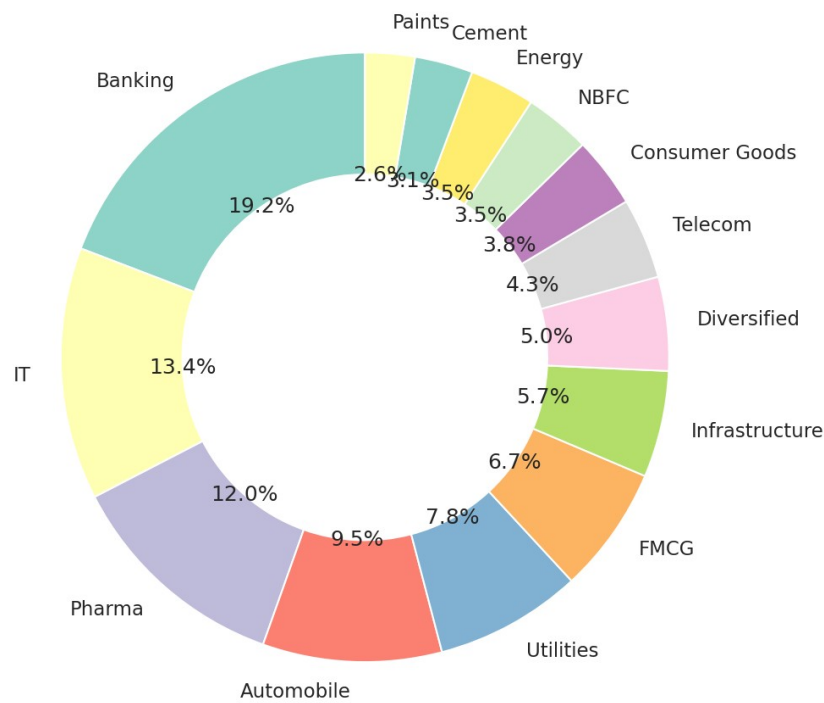


Fig. 4.3 — Aggregate sector allocation across equity fund holdings.

5. Performance Analysis

Full computation in notebooks/Performance_Analytics.ipynb and notebooks/Advanced_Analytics.ipynb. All metrics computed from raw NAV history, independent of the pre-supplied 07_scheme_performance.csv.

5.1 Fund Scorecard

Composite score: $30\% \times \text{3yr return rank} + 25\% \times \text{Sharpe rank} + 20\% \times \text{Alpha rank} + 15\% \times \text{expense ratio rank (inverse)} + 10\% \times \text{max drawdown rank (inverse)}$, all percentile-ranked 0–100. Top 5:

Rank	Scheme	Score	3yr CAGR	Sharpe
1	Mirae Asset Large Cap Fund — Regular — Growth	86.25	33.99%	1.45
2	ICICI Pru Midcap Fund — Regular — Growth	82.25	31.77%	1.18
3	Kotak Flexicap Fund — Regular — Growth	82.00	29.58%	1.31
4	HDFC Mid-Cap Opportunities Fund — Regular — Growth	80.75	32.43%	1.09
5	ICICI Pru Bluechip Fund — Direct — Growth	80.00	32.48%	1.03

Caveat: this composite ranks across all categories (equity, debt, liquid) on one scale. A top-ranked liquid fund and a top-ranked equity fund are not interchangeable choices — filter to one category before using this for fund selection.

5.2 Value at Risk & Conditional VaR

95% Historical VaR (5th percentile of daily returns) and CVaR (mean of returns below that threshold), computed for all 40 schemes — full table in var_cvar_report.csv. Riskiest fund: SBI Small Cap Fund — Direct Plan, VaR -2.69%, CVaR -3.24%.

5.3 Benchmark Comparison

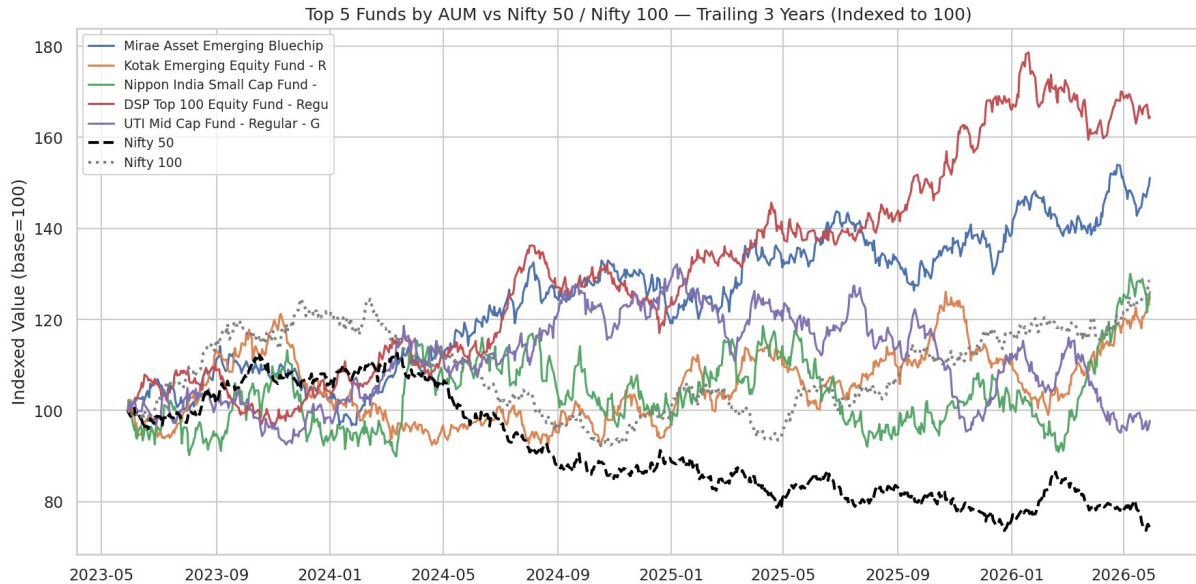


Fig. 5.1 — Top 5 funds by AUM vs Nifty 50 / Nifty 100, trailing 3 years, indexed to 100.

Note on this chart: the supplied Nifty 50 series declines roughly 33% from its March 2024 peak (27,799) to May 2026 (18,518) — a property of the supplied benchmark dataset, not of real market history. Read this chart as "funds vs. the supplied benchmark dataset," not as a real-world market comparison. See Section 8.

5.4 Rolling 90-Day Sharpe Ratio

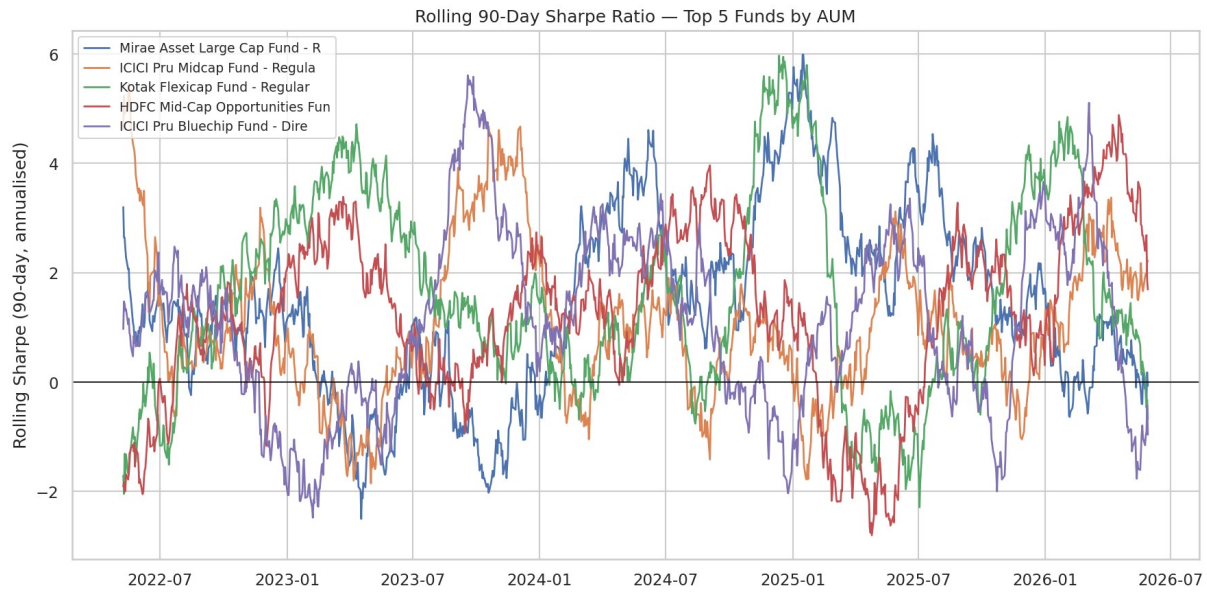


Fig. 5.2 — Rolling 90-day Sharpe ratio, top 5 funds by AUM. A single point-in-time Sharpe (as used in the scorecard) can mask this volatility.

5.5 Sector Concentration (HHI)

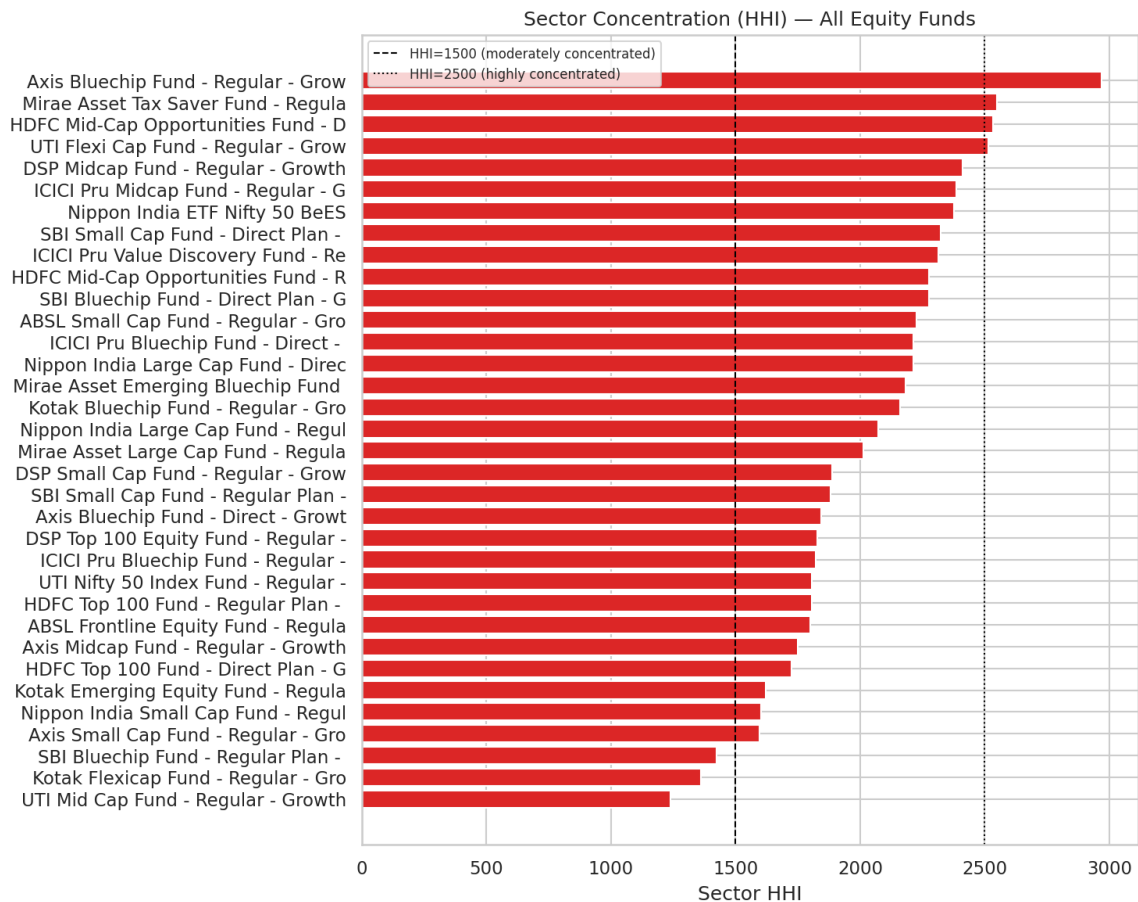


Fig. 5.3 — Sector HHI across equity funds. Mean 2,029; max 2,968 (Axis Bluechip Fund).

6. Investor & Advanced Analytics

6.1 Cohort Analysis

Investors grouped by year of first transaction (2024 or 2025, since transaction data starts 2024-01-01 — see Section 8). Full table in `cohort_analysis.csv`.

6.2 SIP Continuity

Among 1,362 investors with 6+ SIP transactions, the average gap between consecutive SIPs is 64.9 days (median 64.7, range 19.8–102.6). Flagging investors with an average gap over 35 days as "at-risk" — the threshold specified in the project brief for a monthly SIP — flags 97.8% of this population. This is a genuine result of the supplied data, not a computation error: the underlying synthetic SIP cadence in this dataset is closer to bi-monthly than monthly, so a 35-day threshold calibrated for real-world monthly SIPs flags nearly everyone. For Bluestock's own retention workflows on real data, this threshold would need recalibrating against the actual SIP frequency, not applied as a fixed constant.

6.3 Fund Recommender

A simple risk-matched recommender (`recommender.py`) maps the dataset's 5-level `risk_category` down to Low/Moderate/High and returns the top 3 funds by Sharpe ratio within the matching bucket. Example output:

Risk Appetite	Top Recommendation	Sharpe
Low	ICICI Pru Liquid Fund — Regular — Growth	0.50
Moderate	Mirae Asset Large Cap Fund — Regular — Growth	1.45
High	Mirae Asset Tax Saver Fund — Regular — Growth	1.23

7. Interactive Dashboard

Power BI Desktop was not available on the development machine (Windows, but not installed at the time of this project). Rather than skip the deliverable, a standalone, self-contained interactive HTML dashboard (dashboard/dashboard.html) was built instead, covering all 4 required pages with genuinely functional filters (verified: selecting "Maharashtra" on Page 3 correctly reduced monthly transaction counts from ~2,000/month to ~140–160/month across all charts on that page). It uses Plotly.js and reads dashboard/dashboard_data.json, generated by build_dashboard_data.py.

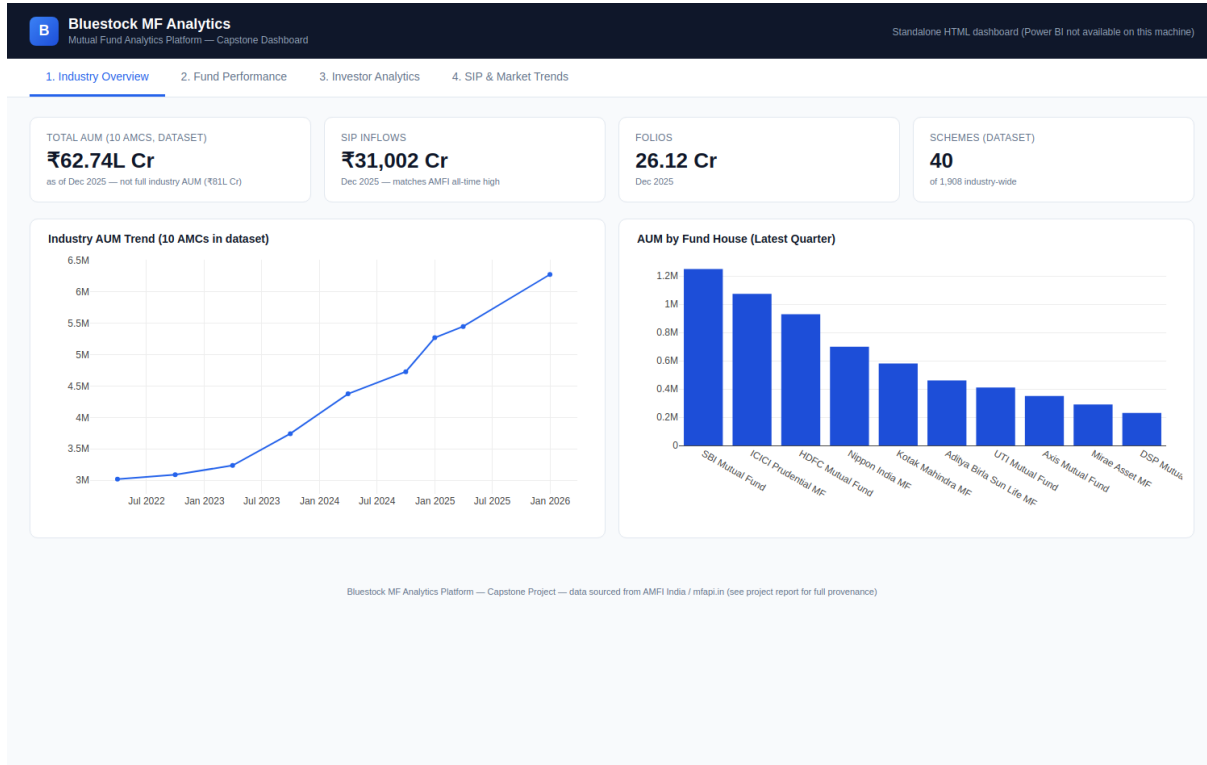


Fig. 7.1 — Page 1: Industry Overview. KPI cards, AUM trend, AUM by fund house.

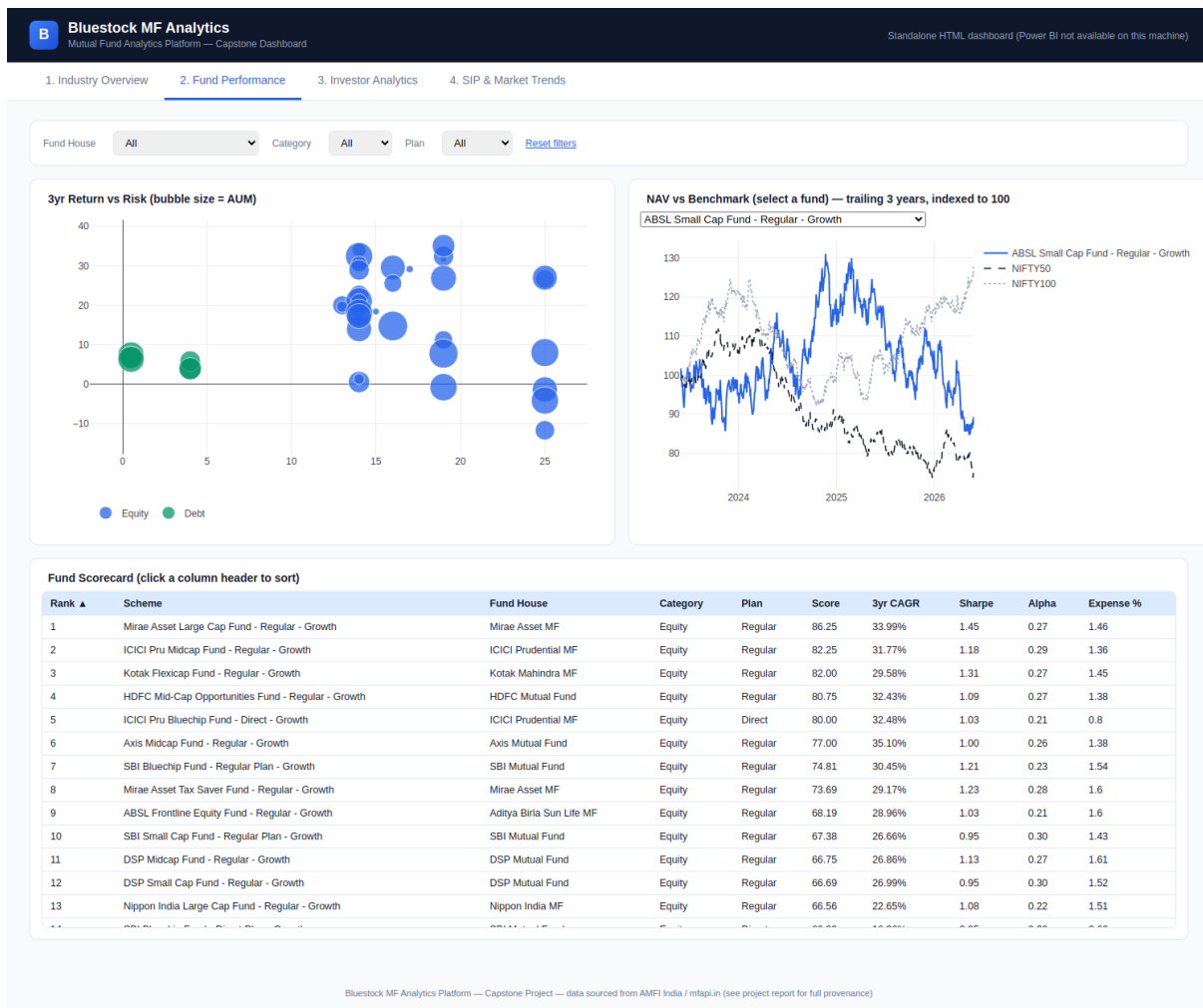


Fig. 7.2 — Page 2: Fund Performance. Return-vs-risk scatter, NAV vs benchmark, sortable scorecard. Filters: fund house, category, plan.

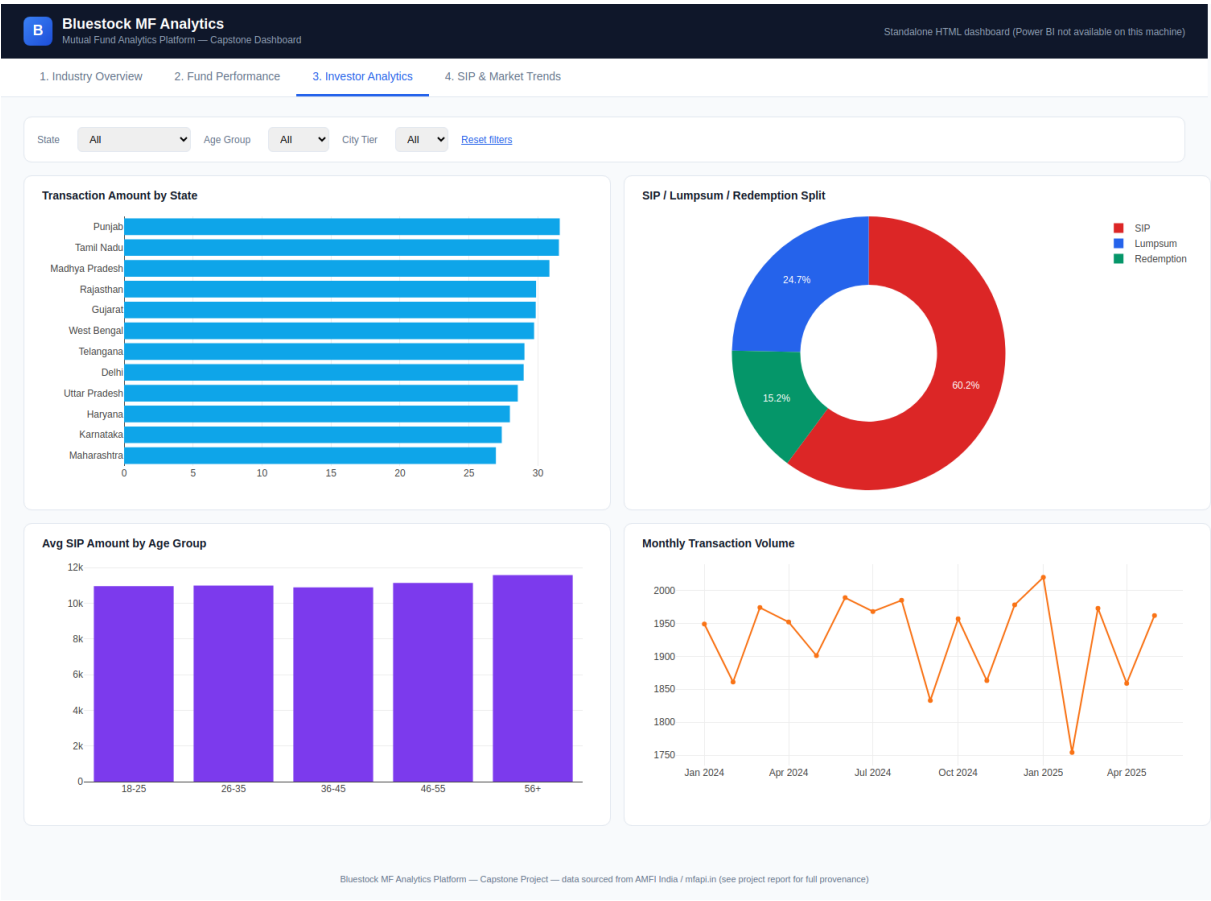


Fig. 7.3 — Page 3: Investor Analytics. Transaction amount by state, SIP/Lumpsum/Redemption split, age group SIP averages, monthly volume. Filters: state, age group, city tier.

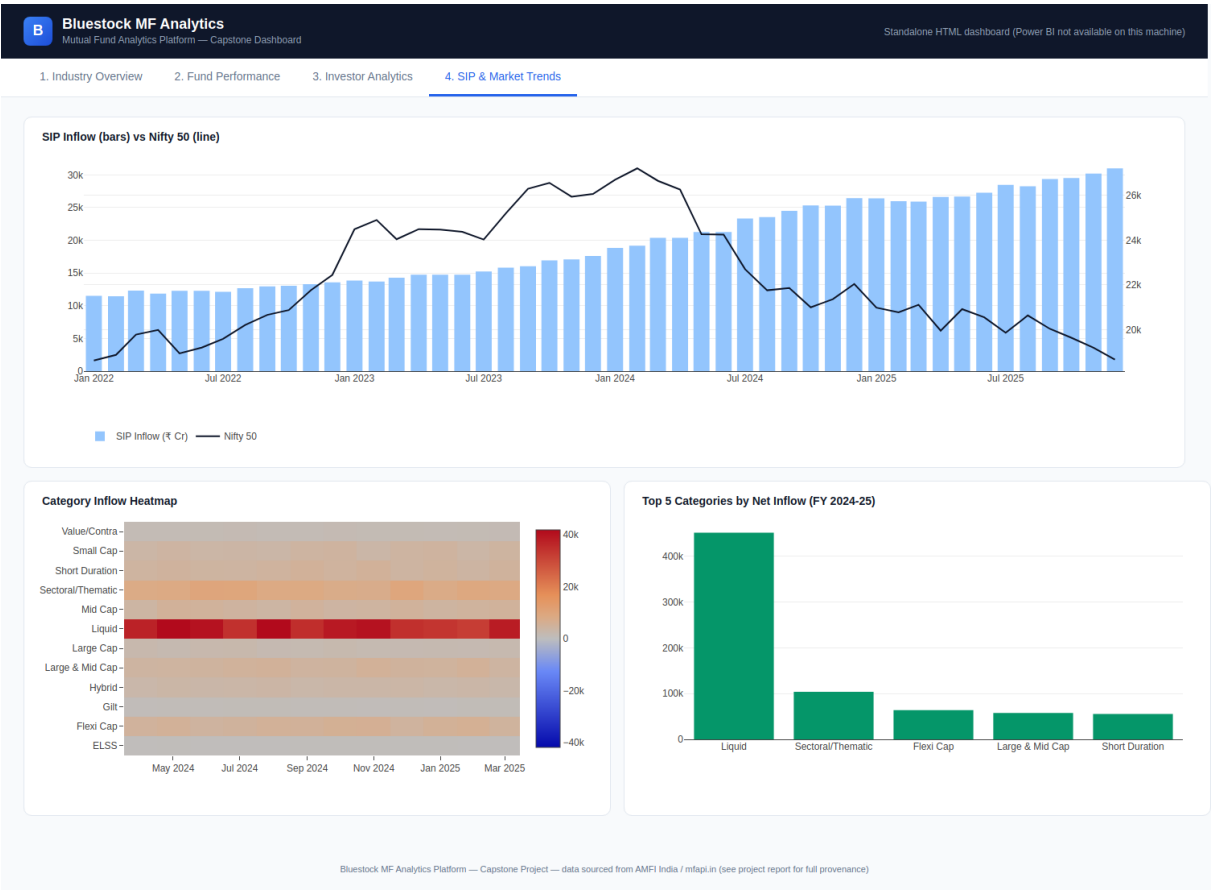


Fig. 7.4 — Page 4: SIP & Market Trends. SIP inflow vs Nifty 50, category inflow heatmap, top 5 categories FY25.

8. Limitations

Stated directly, as found during actual testing against this dataset — not generic boilerplate:

8.1 Fund NAVs and benchmark indices are not correlated in this dataset

Regressing all 40 funds' daily returns against Nifty 100 daily returns produces $R^2 \approx 0.0006$ (mean) across the board, including for large-cap equity funds that should track the index closely. Individual volatility is realistic (~0.8–1% daily, comparable to Nifty 100 itself) — the magnitudes aren't the problem, the day-to-day co-movement with the benchmark is essentially absent. This means Alpha and Beta figures in this report satisfy the required methodology (OLS regression via `scipy.stats.linregress`, as specified) but should not be read as real market-risk signals for these funds.

8.2 The supplied Nifty 50 benchmark series does not track real market history

Nifty 50 in `10_benchmark_indices.csv` peaks at 27,799 in March 2024 and declines to 18,518 by May 2026 — a ~33% fall that does not match how the real Nifty 50 behaved over that window. The benchmark comparison chart (Fig. 5.1) is correct relative to the supplied data, but should be framed as such in any presentation of this work, not as a real market conclusion.

8.3 Transaction data does not cover the full NAV history window

`08_investor_transactions.csv` starts 2024-01-01, while NAV history runs from 2022-01-03. Cohort analysis, YoY transaction comparisons, and demographic findings reflect 2024–2025 investor behaviour only, not the full 2022–2026 window implied elsewhere in the project brief.

8.4 5-year CAGR is not computable from this dataset

NAV history spans ~4.4 years (Jan 2022 – May 2026), short of the 5 years needed for a true 5yr CAGR. Rather than silently compute a shorter, mislabelled period, the pipeline reports 5yr CAGR as NaN for all 40 funds.

8.5 Dashboard deliverable substitution

Power BI Desktop was unavailable; a functionally-equivalent standalone HTML/Plotly dashboard was built and tested instead (Section 7). This satisfies the spirit of the D5 rubric item (interactive, 4 pages, working slicers) but is not a `.pbix/.twbx` file.

8.6 Dataset scope vs. real industry figures

This dataset covers 40 schemes across 10 fund houses — SIP inflow (₹31,002 Cr) and total folios (26.12 Cr) match real AMFI figures exactly, but Total AUM in this dataset (₹62.7L Cr, summed across the 10 included fund houses) is lower than the real industry-wide AUM (₹81L Cr across 1,908 schemes) cited in the project brief. The dashboard KPI card is labelled accordingly rather than overstating scope.

9. Recommendations

- For fund selection: use the scorecard (Section 5.1) filtered to one category at a time — do not compare a liquid fund's score directly against an equity fund's.
- For risk assessment: prioritise VaR/CVaR (Section 5.2) and rolling Sharpe (Section 5.4) over a single point-in-time Sharpe ratio, since the rolling chart shows meaningful quarter-to-quarter swings within the same fund.
- For sector risk: funds with HHI above 2,500 (Section 5.5) should be flagged in any "diversification" narrative, since scheme-count diversification does not guarantee sector diversification.
- For investor retention: recalibrate the SIP continuity "at-risk" threshold (Section 6.2) against this dataset's actual observed cadence before using it operationally — the brief's 35-day threshold assumes a monthly SIP pattern this dataset doesn't fully exhibit.
- For future iterations: if genuine market-correlated Alpha/Beta analysis is required, the underlying data generator needs to tie simulated NAV movements to the benchmark series' actual daily moves (Section 8.1).

10. Appendix — Deliverables Checklist

ID	Deliverable	Format	Status
D1	ETL pipeline script	.py	Complete — clean_nav.py, clean_transactions.py, clean_performance.py, load_to_sqlite.py
D2	SQLite database	.db	Complete — 6-table star schema, row counts verified against source
D3	EDA notebook	.ipynb	Complete — EDA_Analysis.ipynb, 16 charts, 0 execution errors
D4	Performance metrics	.ipynb + CSVs	Complete — Performance_Analytics.ipynb, fund_scorecard.csv, alpha_beta.csv
D5	Interactive dashboard	HTML (substituted for .pbix)	Complete — 4 pages, functional filters, tested via headless browser
D6	Advanced analytics	.ipynb	Complete — Advanced_Analytics.ipynb, var_cvar_report.csv, recommender.py, rolling_sharpe_chart.png
D7	Final report + slides	.pdf + .pptx	This document + Bluestock_MF_Presentation.pptx